

Calcaric Regosols

Description:

Calcaric Regosols are weakly developed mineral soils with limited horizon differentiation, commonly found on young landscapes or recent deposits.

The “Calcaric” qualifier indicates the presence of calcium carbonate (CaCO_3) within the soil profile, usually in the upper 50–100 cm, providing moderate fertility and reducing acidity. These soils form on calcareous parent materials such as limestone, marl, or alluvial deposits and are often light brown to yellowish with low organic matter.

Drainage:

- **Wet season:** Generally well-drained due to coarse texture and low clay content, though local waterlogging may occur in depressions.
- **Dry season:** Soils dry rapidly; moisture retention is low, and surfaces may crust.

Landscape:

Found on gently undulating plains, terraces, colluvial slopes, and alluvial fans in semi-arid to sub-humid regions. Common in parts of Rift Valley and central Kenya on calcareous deposits.

Depth:

Moderately shallow to deep (40–120 cm), with rooting often unrestricted except where stones or calcareous layers are present.

Workability:

- **When wet:** Loose and friable; easy to cultivate.
- **When dry:** Surfaces may become hard and crusted; erosion-prone if left bare.

Nutrient Richness & Cation Exchange Capacity (CEC):

- **CEC:** Low to moderate (5–15 cmol(+)/kg), depending on texture and organic matter.
- **Base saturation:** High (>50%) due to calcium-rich parent material.
- **Nutrients:** Adequate calcium and magnesium; nitrogen and phosphorus often limiting.
- **Organic matter:** Low (<2%).

- **Micronutrients:** Generally adequate; zinc supplementation may be needed.

pH:

Slightly alkaline to neutral (6.5–7.8).

Management:

- Incorporate organic matter (manure, compost) to improve structure and water retention.
- Apply balanced N and P fertilizers to correct deficiencies.
- Use contouring, mulching, and cover crops to reduce erosion.
- Avoid deep tillage on coarse-textured soils.

Suitable Crops:

Cereals (maize, sorghum, millet), legumes (beans, cowpeas), and horticultural crops (tomatoes, onions, cabbage) under proper management.